

Large Language Models Still Can't Plan (2022)

Large Language Models Still Can't Plan — Short Report

1. One-Line Summary

LLMs perform well on popular reasoning benchmarks but fail badly on classical planning tasks that require reasoning about actions, state changes, and goals.

2. Problem

Many researchers claimed that strong performance on GSM8K, CommonsenseQA, and Chain-of-Thought benchmarks indicates reasoning ability. The authors question whether these benchmarks truly measure planning and reasoning about actions.

3. Core Idea

The paper introduces a planning benchmark based on Blocksworld and evaluates whether LLMs can:

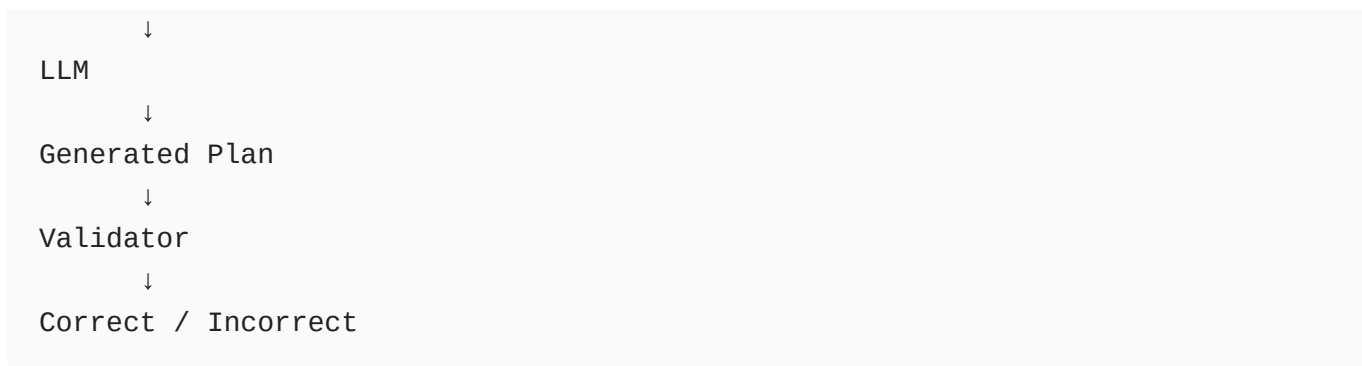
- Generate plans
- Replan after changes
- Execute plans mentally
- Reuse and generalize plans

4. Key Concepts

- **Planning Problem:** Initial State + Goal + Actions
- **State:** Current facts about the world
- **Action:** Operation that changes the state
- **Preconditions:** Conditions required for an action
- **Effects:** Changes produced by an action
- **Blocksworld:** Benchmark domain used for evaluation

5. Method Pipeline

```
Planning Domain
    ↓
Problem Generator
    ↓
Natural Language Prompt
```



6. Results

Very Poor Performance

- Plan Generation: GPT-3 $\approx$ 0.6%
- Optimal Planning: GPT-3 $\approx$ 0.2%
- InstructGPT improves slightly but remains poor.

Moderate Performance

- Replanning
- Plan Generalization
- Plan Reuse

These tasks perform better than plan generation but remain far from human performance.

Strong Performance

Goal reformulation and goal recognition tasks achieve roughly 70–77% accuracy.

This is the paper's most interesting finding:

LLMs can often identify equivalent goals and manipulate plan descriptions without being able to construct valid plans.

7. Weaknesses

- Uses only Blocksworld.
- Strong dependence on prompt design.
- Evaluates pre-GPT-4 models.
- Planning is only one aspect of reasoning..

Final Takeaway

Success on reasoning benchmarks does not imply planning ability. GPT-3 and BLOOM struggled with planning tasks, suggesting that next-token prediction alone is insufficient for

robust classical planning. The paper's strongest contribution is showing that planning should be evaluated separately from arithmetic and commonsense reasoning. LLMs appear much better at pattern completion than explicit state-space search.